

Notes: Loughran & McDonald (JF, 2011)

When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks

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1 Big picture

- **Question.** Can standard generic sentiment dictionaries measure tone in financial text, especially 10-K filings?
- **Core critique.** The widely used Harvard-IV-4 TagNeg dictionary misclassifies many common financial words as negative, including words such as *tax*, *cost*, *capital*, *liability*, and *board*.
- **Main fact.** Almost three-fourths of Harvard negative word counts in 10-Ks come from words that are typically not negative in a financial context.
- **Contribution.** The authors build finance-specific word lists, especially a financial negative list, and compare it with the Harvard negative list.
- **Data.** The sample contains 50,115 firm-year 10-K filings from 1994 to 2008, plus a smaller MD&A subsample.
- **Main return result.** Negative words from the finance-specific list are related to filing-window excess returns. Harvard negative words are much weaker when simple proportional counts are used.
- **Term weighting.** When term-frequency/inverse-document-frequency weighting is used, the gap between Harvard and financial lists narrows because very common misclassified words receive less weight.
- **Additional word lists.** The paper also constructs positive, uncertainty, litigious, strong modal, and weak modal lists.
- **Validation.** The word lists are related to filing returns, trading volume, return volatility, fraud, internal control weaknesses, and standardized unexpected earnings.
- **Takeaway.** Textual analysis depends critically on context-specific dictionaries. A word list designed for psychology and sociology can generate noisy or biased finance measures.

2 Data and measurement

2.1 10-K sample construction

- The starting universe contains EDGAR 10-K and 10-K405 filings from 1994 to 2008.
- The authors exclude amended documents and repeated annual filings.
- Filings are matched to CRSP and Compustat.
- Firms must have sufficient returns and volume data, valid market capitalization, and at least 2,000 words in the filing.
- The final 10-K sample contains 50,115 firm-year filings from 8,341 unique firms.
- The MD&A sample contains 37,287 filings with at least 250 words in the MD&A section.

Interpretation: The sample filters ensure that each text observation can be linked to market reactions and firm characteristics. The MD&A subsample asks whether focusing on the narrative section improves tone measurement.

2.2 Word lists

The paper compares the Harvard-IV-4 negative list with finance-specific lists:

- **H4N-Inf:** Harvard-IV-4 TagNeg words plus inflections.
- **Fin-Neg:** finance-specific negative words developed by the authors.

- **Fin-Pos:** finance-specific positive words.
- **Fin-Unc:** uncertainty words.
- **Fin-Lit:** litigious words.
- **MW-Strong:** strong modal words.
- **MW-Weak:** weak modal words.

Interpretation: The central methodological claim is that financial text has its own dialect. A word can be negative in ordinary language but neutral in a 10-K.

2.3 Proportional word counts

For document i and dictionary D , the proportional measure is:

$$Prop_{i,D} = \frac{\sum_{w \in D} Count_{i,w}}{TotalWords_i}. \quad (1)$$

Interpretation: This is the standard bag-of-words approach. It is easy to compute but vulnerable to high-frequency contextually neutral words.

2.4 Term weighting

The paper also uses a term-frequency/inverse-document-frequency style weighting. Conceptually,

$$TFIDF_{i,w} = TF_{i,w} \times IDF_w, \quad (2)$$

where TF dampens within-document word frequency and IDF downweights words that appear in many documents.

Interpretation: Term weighting reduces the influence of extremely common words, such as *taxes* or *costs*, that are not very informative about tone.

2.5 Market and accounting outcomes

The paper validates word lists using several outcomes:

- filing-window excess return from days 0 through 3 after the 10-K filing;
- filing-window abnormal volume;
- postevent return volatility;
- shareholder class action lawsuits for accounting issues;
- material weaknesses in internal controls;
- standardized unexpected earnings.

Interpretation: A useful tone measure should be related to economically meaningful outcomes, not just dictionary counts.

3 Models and empirical specifications

3.1 Filing-period return regression

The main return regression is of the form

$$ExcessReturn_i = \alpha + \beta_1 Tone_i + \gamma' Controls_i + IndustryFE_i + \epsilon_i. \quad (3)$$

Controls include log size, log book-to-market, log share turnover, pre-filing Fama–French alpha, institutional ownership, NASDAQ dummy, and industry dummies.

Interpretation: A useful negative tone measure should have a negative coefficient: more negative language should be associated with lower filing-period excess returns.

3.2 MD&A regression

The MD&A tests replace the full-document tone measure with tone computed only from the MD&A section:

$$ExcessReturn_i = \alpha + \beta_1 Tone_i^{MD\&A} + \gamma' Controls_i + IndustryFE_i + \epsilon_i. \quad (4)$$

Interpretation: This asks whether the MD&A section gives a cleaner tone signal than the entire 10-K.

3.3 Additional word-list regressions

For the extended word lists, the dependent variables are event returns, event abnormal volume, and postevent return volatility:

$$Y_i = \alpha + \sum_D \beta_D Tone_{i,D} + \gamma' Controls_i + IndustryFE_i + \epsilon_i. \quad (5)$$

Interpretation: Different dictionaries are designed to capture different aspects of disclosure tone and risk. For example, uncertainty and litigious words should be related to volatility and litigation outcomes.

3.4 Fraud and material weakness logit regressions

For fraud and internal-control weakness outcomes, the paper estimates logit models:

$$Pr(Event_i = 1) = \Lambda \left(\alpha + \sum_D \beta_D Tone_{i,D} + \gamma' Controls_i \right). \quad (6)$$

Interpretation: These regressions test whether dictionary measures forecast or correlate with legally and accounting-relevant bad outcomes.

3.5 Standardized unexpected earnings

The paper also estimates regressions of standardized unexpected earnings on tone:

$$SUE_i = \alpha + \beta_1 Tone_i + \gamma' Controls_i + \epsilon_i. \quad (7)$$

Interpretation: Negative language can be forward-looking or strategic. If managers lower expectations through negative disclosure, negative words may be related to subsequent earnings surprises.

4 Identification and empirical design

4.1 What identifies the dictionary problem

- The authors compare which words the Harvard list classifies as negative against actual usage in 10-Ks.
- They show that many high-frequency Harvard negative words are ordinary financial vocabulary.
- They test whether different word lists produce different relations with market reactions.

Interpretation: The identification is conceptual and empirical: the paper shows both direct misclassification and worse performance in return regressions.

4.2 Why term weighting matters

- High-frequency words dominate proportional counts.
- If high-frequency words are misclassified, they add noise and possibly bias.
- TF-IDF dampens common terms, making the Harvard and Fin-Neg lists more comparable.

Interpretation: The method can partly compensate for a poor dictionary, but a finance-specific dictionary remains more interpretable.

4.3 What the paper cannot fully solve

- Words are polysemous; no word list can perfectly classify tone in every sentence.
- The analysis is mostly bag-of-words and does not fully model negation, syntax, or context.
- The word lists are designed for 10-Ks and may not transfer perfectly to other financial texts.
- Regression relations do not prove that words cause returns, fraud, or earnings outcomes.

Interpretation: The paper's core contribution is not perfect sentiment analysis. It shows that domain-specific dictionaries are essential for credible finance applications.

5 Tables and figures

The figures and tables below are directly cropped from the paper and arranged compactly.

5.1 Table I and Table II: sample construction and summary statistics

Read:

- Table I shows how the sample falls from 121,217 filings to 50,115 final firm-year 10-Ks.
- The largest reduction comes from requiring a CRSP PERMNO match.
- The MD&A subsample contains 37,287 filings.
- Table II shows that the Harvard negative list is much more frequent than the finance-specific negative list.
- Full 10-Ks average 3.79% H4N-Inf words but only 1.39% Fin-Neg words.

Economic message: The paper starts by showing that the difference between general and financial dictionaries is large in raw text measurement.

10-K Sample Creation

This table reports the impact of various data filters on initial 10-K sample size.

Source/Filter	Sample Size	Observations Removed
<i>Full 10-K Document</i>		
EDGAR 10-K/10-K405 1994–2008 complete sample (excluding duplicates)	121,217	
Include only first filing in a given year	120,290	927
At least 180 days between a given firm's 10-K filings	120,074	216
CRSP PERMNO match	75,252	44,822
Reported on CRSP as an ordinary common equity firm	70,061	5,191
CRSP market capitalization data available	64,227	5,834
Price on filing date day minus one $\geq$ \$3	55,946	8,281
Returns and volume for day 0–3 event period	55,630	316
NYSE, AMEX, or Nasdaq exchange listing	55,612	18
At least 60 days of returns and volume in year prior to and following file date	55,038	574
Book-to-market COMPUSTAT data available and book value > 0	50,268	4,770
Number of words in 10-K $\geq$ 2,000	50,115	153
<i>Firm-Year Sample</i>		
Number of unique firms	8,341	
Average number of years per firm	6	
<i>Management Discussion and Analysis (MD&A)</i>		
<i>Subsection</i>		
Subset of 10-K sample where MD&A section could be identified	49,179	936
MD&A section $\geq$ 250 words	37,287	11,892

II. Data, Variables, and Term Weights

A. The 10-K Sample

Table I: 10-K sample creation

Summary Statistics for the 1994 to 2008 10-K Sample

The first seven variables represent the proportion of occurrences for a given word list relative to the total number of words. The word lists are available in the Internet Appendix or at <http://www.nd.edu/~mcdonald/Word.Lists.html>. See the Appendix for the other variable definitions. The sample sizes for the last three earnings-related variables are 28,679 for the full 10-K sample and 21,240 for the MD&A subsample.

Variable	Full 10-K Document (N = 50,115)			MD&A Section (N = 37,287)		
	Mean	Median	Standard Deviation	Mean	Median	Standard Deviation
<i>Word Lists</i>						
H4N-Inf (H4N w/ infections)	3.79%	3.84%	0.76%	4.83%	4.79%	0.89%
Fin-Neg (negative)	1.39%	1.36%	0.55%	1.51%	1.43%	0.67%
Fin-Pos (positive)	0.75%	0.74%	0.21%	0.83%	0.79%	0.32%
Fin-Unc (uncertainty)	1.20%	1.20%	0.32%	1.56%	1.48%	0.62%
Fin-Lit (litigious)	1.10%	0.95%	0.53%	0.60%	0.51%	0.43%
MW-Strong (strong modal words)	0.26%	0.24%	0.11%	0.30%	0.27%	0.17%
MW-Weak (weak modal words)	0.43%	0.39%	0.21%	0.43%	0.34%	0.32%
<i>Other Variables</i>						
Event period [0,3] excess return	−0.12%	−0.19%	6.82%	−0.23%	−0.28%	7.26%
Size (\$billions)	\$3.09	\$0.33	\$14.94	\$2.12	\$0.30	\$9.62
Book-to-market	0.613	0.512	0.459	0.611	0.501	0.477
Turnover	1.519	0.947	2.295	1.695	1.104	2.508
One-year preevent FF alpha	0.07%	0.04%	0.20%	0.07%	0.05%	0.21%
Institutional ownership	48.34%	48.07%	28.66%	49.23%	48.52%	29.33%
NASDAQ dummy	56.15%	100.00%	49.62%	60.12%	100.00%	48.97%
Standardized unexpected earnings	−0.02%	0.03%	0.76%	−0.03%	0.03%	0.82%
Analysts' earnings forecast dispersion	0.17%	0.07%	0.33%	0.19%	0.08%	0.36%
Analysts' earnings revisions	−0.21%	−0.04%	0.69%	−0.24%	−0.05%	0.74%

the mean for H4N-Inf rising from about 3.5% to 4.3% and the mean for Fin-

Table II: summary statistics

5.2 Table III: common Harvard and financial negative words

Read:

- Panel A shows that high-frequency Harvard negative words include terms such as *tax*, *costs*, *capital*, *liabilities*, *foreign*, and *vice*.
- Many of these are not typically negative in financial reporting.
- Panel B shows the most frequent Fin-Neg words, such as *loss*, *losses*, *claims*, *impairment*, *adverse*, *restated*, and *litigation*.
- The Fin-Neg list better maps to financial bad news.

Economic message: This table is the paper's core diagnostic. The Harvard list mistakes ordinary business vocabulary for negative tone.

5.3 Figure 1 and Table IV: filing returns and negative word lists

Read:

- Figure 1 sorts firms by negative word proportions.
- The Fin-Neg list shows a monotonic pattern: higher negative tone corresponds to lower median filing-period excess returns.
- The Harvard list does not show the same monotonic pattern.
- Table IV confirms this in regressions. Fin-Neg is significantly negative under proportional weights, while H4N-Inf is not.
- With TF-IDF weights, both lists become significant because common misclassified words are downweighted.

Thirty Most Frequent Words Occurring in 10-Ks from the H4N-Inf and Fin-Neg Word Lists

The H4N-Inf word list is based on the Harvard-IV-4 Psychosociological Dictionary TagNeg file. We extend the original word list to include appropriate inflections. The Fin-Neg word list includes negative words from a list of all words occurring in the full sample of 10-Ks filed over 1994 to 2008. The word lists are available in the Internet Appendix or at http://www.nd.edu/~mcdonald/Word_Lists.html. There are 4,187 H4N-Inf words, based on 2,005 root words in the original H4N list. Fin-Neg consists of 2,337 words, including inflections. The results in this table are based on the sample of 50,115 10-Ks with complete data for our regression variables downloaded from EDGAR for the period 1994 to 2008. The MD&A subsample comprises 37,287 observations. Results are presented for the full 10-K documents and the corresponding MD&A portion of the 10-Ks. Words not appearing in both the full 10-K and MD&A subsample lists are italicized.

Panel A: H4N-Inf							
Full 10-K Document				MD&A Subsection			
Word in Fin-Neg	Word	% of Total Fin-Neg Word Count	Cumulative %	Word in Fin-Neg	Word	% of Total Fin-Neg Word Count	Cumulative %
	TAX	4.83%	4.83%		COSTS	6.45%	6.45%
	COSTS	4.61%	9.44%		EXPENSES	5.51%	11.96%
✓	LOSS	3.77%	13.21%		EXPENSE	4.70%	16.66%
	CAPITAL	3.62%	16.83%		TAX	4.68%	21.34%
	COST	3.51%	20.34%		CAPITAL	4.24%	25.58%
	EXPENSE	3.12%	23.46%		COST	3.70%	29.28%
	EXPENSES	2.92%	26.38%	✓	LOSS	3.29%	32.57%
	LIABILITIES	2.66%	29.04%		DECREASE	3.06%	35.63%
	SERVICE	2.57%	31.61%		RISK	2.97%	38.60%
	RISK	2.34%	33.95%	✓	LOSSES	2.62%	41.22%
	TAXES	2.23%	36.18%		DECREASED	2.21%	43.44%
✓	LOSSES	2.20%	38.38%		LIABILITIES	2.15%	45.58%
	BOARD	2.13%	40.51%		LOWER	2.10%	47.69%
	FOREIGN	1.68%	42.20%		TAXES	1.95%	49.63%
	VICE	1.52%	43.71%		SERVICE	1.91%	51.55%
	LIABILITY	1.41%	45.12%		FOREIGN	1.87%	53.42%
	DECREASE	1.29%	46.41%	✓	IMPAIRMENT	1.63%	55.05%
✓	IMPAIRMENT	1.18%	47.59%		CHARGES	1.40%	56.44%
	LIMITED	1.10%	48.69%		LIABILITY	1.16%	57.60%
	LOWER	1.01%	49.70%		CHARGE	1.16%	58.76%
✓	AGAINST	1.00%	50.70%		RISKS	1.05%	59.80%
	MATTERS	0.99%	51.69%	✓	DECLINE	1.00%	60.80%
✓	ADVERSE	0.94%	52.63%		DEPRECIATION	0.92%	61.72%
	CHARGES	0.94%	53.57%		MAKE	0.86%	62.58%
	MAKE	0.89%	54.46%	✓	ADVERSE	0.84%	63.42%
	ORDER	0.88%	55.33%		BOARD	0.79%	64.21%
	RISKS	0.85%	56.19%		LIMITED	0.78%	64.99%
	DEPRECIATION	0.85%	57.04%		EXCESS	0.71%	65.70%
	CHARGE	0.83%	57.87%		ORDER	0.70%	66.40%
	EXCESS	0.82%	58.69%	✓	AGAINST	0.70%	67.10%

(continued)

Table III—Continued

Panel B: Fin-Neg							
Full 10-K Document				MD&A Subsection			
Word in H4N-Inf	Word	% of Total Fin-Neg Word Count	Cumulative %	Word in H4N-Inf	Word	% of Total Fin-Neg Word Count	Cumulative %
✓	LOSS	9.73%	9.73%	✓	LOSS	9.51%	9.51%
✓	LOSSES	5.67%	15.40%	✓	LOSSES	7.58%	17.10%
✓	CLAIMS	3.15%	18.55%	✓	IMPAIRMENT	4.71%	21.81%
✓	IMPAIRMENT	3.04%	21.59%	✓	RESTRUCTURING	2.93%	24.74%
✓	AGAINST	2.58%	24.17%	✓	DECLINE	2.89%	27.62%
✓	ADVERSE	2.44%	26.61%		CLAIMS	2.71%	30.33%
✓	RESTATE	2.09%	28.70%	✓	ADVERSE	2.44%	32.77%
✓	ADVERSELY	1.75%	30.45%	✓	AGAINST	2.01%	34.78%
✓	RESTRUCTURING	1.72%	32.17%	✓	ADVERSELY	1.94%	36.72%
	LITIGATION	1.67%	33.83%		LITIGATION	1.67%	38.40%
	DISCONTINUED	1.57%	35.40%		CRITICAL	1.63%	40.03%
	TERMINATION	1.35%	36.75%		DISCONTINUED	1.62%	41.64%
✓	DECLINE	1.19%	37.93%	✓	DECLINED	1.30%	42.94%
✓	CLOSING	1.08%	39.01%		TERMINATION	1.06%	44.00%
✓	FAILURE	0.97%	39.98%	✓	NEGATIVE	0.96%	44.96%
✓	UNABLE	0.84%	40.82%	✓	FAILURE	0.93%	45.89%
✓	DAMAGES	0.82%	41.64%		UNABLE	0.91%	46.80%
✓	DOUBTFUL	0.77%	42.41%	✓	CLOSING	0.86%	47.65%
✓	LIMITATIONS	0.75%	43.17%		NONPERFORMING	0.81%	48.47%
✓	FORCE	0.74%	43.91%	✓	IMPAIRED	0.81%	49.28%
✓	VOLATILITY	0.73%	44.64%	✓	VOLATILITY	0.79%	50.07%
	CRITICAL	0.73%	45.37%	✓	FORCE	0.75%	50.82%
✓	IMPAIRED	0.70%	46.07%	✓	NEGATIVELY	0.73%	51.56%
✓	TERMINATED	0.677%	46.77%	✓	DOUBTFUL	0.72%	52.27%
✓	COMPLAINT	0.63%	47.39%	✓	CLOSED	0.70%	52.97%
✓	DEFAULT	0.57%	47.96%	✓	DIFFICULT	0.69%	53.66%
✓	NEGATIVE	0.51%	48.47%	✓	DECLINES	0.63%	54.29%
✓	DEFENDANTS	0.51%	48.99%	✓	EXPOSED	0.60%	54.89%
✓	PLAINTIFFS	0.51%	49.49%	✓	DEFAULT	0.59%	55.48%
✓	DIFFICULT	0.50%	50.00%	✓	DELAYS	0.56%	56.04%

typically negative in the context of financial reporting. If more than the top 25 words are examined, we find that almost three-fourths (73.8%) of the Harvard negative word count typically does not have negative meaning in financial documents based on our classification.

Economic message: The finance-specific negative list performs better because it captures economically relevant negative tone rather than generic negative vocabulary.

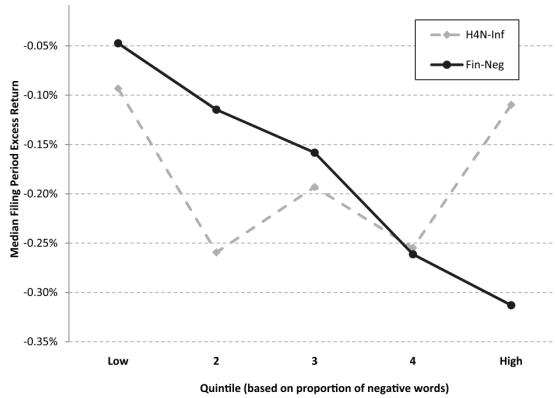


Figure 1. Median filing period excess return by quintile for the Harvard-IV-4 Psychosociological Dictionary TagNeg word list with inflections (H4N-Inf) and the Financial Negative (Fin-Neg) word list. For each of the word lists, the sample of 50,115 10-Ks is divided into five portfolios based on the proportion of negative words. The filing period return is the holding period excess return for the 10-K file date through the subsequent 3 days, where the excess is

Figure 1: median filing-period excess returns by quintile

Comparison of Negative Word Lists Using Filing Period Excess Return Regressions

The dependent variable in each regression is the event period excess return (defined as the firm's buy-and-hold stock return minus the CRSP value-weighted buy-and-hold market index return over the 4-day event window, expressed as a percent). The proportional weights are the word list counts relative to the total number of words appearing in a firm's 10-K. The tf.idf weighted values are defined in equation (1) of the text. See the Appendix for the other variable definitions. Fama-French (1997) industry dummies (based on 48 industries) and a constant are also included in each regression. The coefficients are based on 60 quarterly Fama-MacBeth (1973) regressions with Newey-West standard (1987) errors using one lag. The estimates use a sample of 50,115 10-Ks over 1994 to 2008.

	Proportional Weights		tf.idf Weights	
	(1)	(2)	(3)	(4)
Word Lists				
H4N-Inf (Harvard-IV-4-Neg with inflections)	-7.422 (-1.35)		-0.003 (-3.16)	
Fin-Neg (negative)		-19.538 (-2.64)		-0.003 (-3.11)
Control Variables				
Log(size)	0.123 (2.87)	0.127 (2.93)	0.131 (2.96)	0.132 (2.97)
Log(book-to-market)	0.279 (3.35)	0.280 (3.45)	0.273 (3.37)	0.277 (3.41)
Log(share turnover)	-0.284 (-2.46)	-0.269 (-2.36)	-0.254 (-2.32)	-0.255 (-2.31)
Pre.FFalpha	-2.500 (-0.06)	-3.861 (-0.09)	-5.319 (-0.12)	-6.081 (-0.14)
Institutional ownership	0.278 (0.93)	0.261 (0.86)	0.254 (0.87)	0.255 (0.87)
NASDAQ dummy	0.073 (0.86)	0.073 (0.87)	0.083 (0.97)	0.080 (0.94)
Average R ²	2.44%	2.52%	2.64%	2.63%

share turnover, prefile date Fama-French alpha, institutional ownership, and a NASDAQ dummy. Included in each regression are Fama and French (1997) 48-industry dummies and a constant. Because of the panel structure of the data, we use a Fama-MacBeth (1973) methodology where the firms are grouped by

Table IV: filing-period excess return regressions

5.4 Table V: MD&A-only return regressions

Read:

- Table V repeats the return regression using only the MD&A section.
- The proportional MD&A measures are not statistically significant.
- The TF-IDF weighted measures are only marginally significant.
- The MD&A section does not give a clearly stronger tone measure than the full 10-K.

Economic message: It is not always better to focus narrowly on the MD&A. Full-document tone can contain useful information.

Comparison of Negative Word Lists Using Filing Period Excess Return Regressions: MD&A Section

The sample is now based on the MD&A section of 10-Ks over 1994 to 2008, where the MD&A section contains at least 250 words ($N = 37,287$). The dependent variable in each regression is the event period excess return (defined as the firm's buy-and-hold stock return minus the CRSP value-weighted buy-and-hold market index return over the 4-day event window, expressed as a percent). The proportional weights are the word list counts relative to the total number of words appearing in a firm's 10-K. The tf.idf weighted values are defined in equation (1) of the text. See the Appendix for the other variable definitions. Fama-French (1997) industry dummies (based on 48 industries) and a constant are also included in each regression. The coefficients are based on 60 quarterly Fama-MacBeth (1973) regressions with Newey-West (1987) standard errors using one lag.

	Proportional Weights		tf.idf Weights	
	(1)	(2)	(3)	(4)
Word Lists				
H4N-Inf (only MD&A)	1.892 (0.35)		-0.005 (-1.96)	
Fin-Neg (only MD&A)		-5.344 (-0.68)		-0.006 (-1.96)
Control Variables				
Log(size)	0.157 (3.06)	0.162 (3.10)	0.172 (3.32)	0.172 (3.28)
Log(book-to-market)	0.324 (3.57)	0.330 (3.59)	0.334 (3.64)	0.335 (3.65)
Log(share turnover)	-0.364 (-2.84)	-0.362 (-2.82)	-0.345 (-2.79)	-0.341 (-2.76)
Pre_FFAlpha	-21.977 (-0.45)	-22.279 (-0.45)	-23.050 (-0.47)	-23.168 (-0.47)
Institutional ownership	0.271 (0.82)	0.264 (0.79)	0.244 (0.74)	0.245 (0.74)
NASDAQ dummy	0.146 (1.45)	0.144 (1.39)	0.141 (1.40)	0.139 (1.38)
Average R^2	2.45%	2.70%	2.65%	2.76%

included. The results for both word lists using proportional measures are not significant and are only marginally significant for the weighted measures. The results of this table indicate that the MD&A section does not contain higher

**Table VI
Additional Word Lists, Filing Period Returns, Filing Period Abnormal Volume, and Postevent Return Volatility**

The table reports the coefficients for the tone-related variables. Each entry in the table is based on a separate Fama-MacBeth (1973) regression (i.e., 42 different Fama-MacBeth regressions). The event period excess return is defined as the firm's buy-and-hold stock return minus the CRSP value-weighted buy-and-hold market index return over the 4-day event window, expressed as a percent. Event period abnormal volume is the sum of volumes for the 4-day filing event period, where volume is standardized based on the 60 days prior to the file date. Postevent return volatility is the root-mean square error from a Fama-French (1990) regression on daily returns data for the 202 days following the file date, with the first 5 days following the file date excluded. The coefficients are based on 60 (59 for Postevent return volatility) quarterly Fama-MacBeth (1973) regressions with Newey-West (1987) standard errors using one lag. Each coefficient reported is from a regression also containing the control variables appearing in Table IV (see the Appendix for definitions), Fama-French (1997) industry dummies (based on 48 industries), and a constant. The estimates are based on the sample of 50,115 (49,179 for Postevent return volatility) 10-Ks over 1994 to 2008.

Dependent Variable	H4N-Inf	Finance Dictionaries					
		Negative	Positive	Uncertainty	Litigious	Modal Strong	Modal Weak
Panel A: Proportional Weights							
Event period excess return	-7.422 (-1.35)	-19.538 (-2.64)	-21.696 (-1.18)	-42.026 (-4.13)	9.705 (1.17)	-149.658 (-3.82)	-60.230 (-2.43)
Event period abnormal volume (coefficient /100)	2.735 (2.02)	6.453 (3.11)	-1.957 (-0.20)	2.220 (0.48)	0.057 (0.02)	21.430 (1.67)	4.300 (0.74)
Postevent return volatility	11.336 (8.59)	34.337 (12.59)	18.803 (3.47)	33.973 (8.34)	-0.299 (-0.23)	152.312 (12.32)	59.239 (8.58)
Panel B: tf.idf Weights							
Event period excess return	-0.003 (-3.16)	-0.003 (-3.11)	-0.011 (-2.27)	-0.022 (-4.04)	-0.001 (-0.62)	-0.065 (-2.28)	-0.080 (-3.44)
Event period abnormal volume	0.086 (4.30)	0.098 (4.40)	0.159 (1.03)	0.409 (2.50)	0.135 (2.60)	0.046 (0.03)	0.864 (1.21)
Postevent return volatility	0.004 (12.91)	0.004 (11.87)	0.014 (12.52)	0.020 (8.95)	0.006 (10.10)	0.073 (7.47)	0.069 (8.21)

5.5 Table VI: additional word lists, volume, and volatility

Read:

- Table VI studies event returns, event abnormal volume, and postevent return volatility.
- Negative words are associated with lower event returns.
- Positive words are associated with higher event returns in some specifications.
- Uncertainty and litigious words are strongly related to abnormal volume and postevent volatility.
- Modal strong and modal weak words also contain information about future uncertainty and volatility.

Economic message: Textual tone is multidimensional. Negative words are not the only useful category; uncertainty and litigation vocabulary are especially informative about risk.

5.6 Table VII and Table VIII: fraud, internal controls, and earnings surprises

Read:

- Table VII relates word lists to accounting fraud and material weaknesses in internal controls.
- Fin-Neg and modal strong words are positively related to material weaknesses.
- The TF-IDF weighted lists generally perform better than proportional lists.
- Table VIII shows that negative tone is positively related to standardized unexpected earnings in later quarters.
- This suggests that managers may use negative language partly to lower expectations.

Economic message: Finance-specific textual measures have information beyond stock returns. They are related to litigation, accounting problems, and future earnings outcomes.

Logit Regressions for Shareholder Class Action Suits for Accounting Issues and Self-reported Material Weakness in Internal Controls

The table reports the coefficients for the tone-related variables. Each entry in the table is based on a separate logit regression (i.e., 28 separate regressions). The binary dependent variable for the logit regression when the dependent variable is Fraud is equal to one if a firm had a 10b-5 class action lawsuit filed in the year after the 10-K filing date or if the 10-K file date falls within the purported violation period reported in the 10b-5 ($N = 585$). The 10b-5 sample includes only firms accused of accounting fraud. To parallel the dates available in the 10b-5 sample, all 10-Ks from 1/1/1994 through 9/23/2004 are included in the sample ($N = 35,992$). For the dependent variable labeled Material weakness, the binary dependent variable is set equal to one if within 18 months of the 10-K file date a disclosure of material weakness is reported in a subsequent 10-K, 10-Q, or 8-K. The material weakness sample is taken from Doyle, Ge, and McVay (2007). To parallel the dates of their data set, the observation period for this second sample is all 10-Ks from January 1, 2001 through October 30, 2005. The total material weakness sample is 17,143 with 708 cases indicating a material weakness event. For all regressions, the standard errors used to calculate the z -statistics, in parentheses, are corrected for rare-event bias using the method of Tomz, King, and Zeng (2003). The control variables from Table IV, Fama-French five industry dummies, year dummies, and a constant are also included in the logit regressions. See the Appendix for the control variable definitions.

Finance Dictionaries							
Dependent Variable	H4N-Inf	Negative	Positive	Uncertainty	Litigious	Modal Strong	Modal Weak
Panel A: Proportional Weights							
Fraud	3.109 (0.52)	9.207 (1.20)	-6.031 (-0.34)	19.425 (1.42)	-0.003 (-0.00)	1.066 (0.03)	-45.369 (-1.94)
Material weakness	9.082 (1.43)	31.342 (3.95)	-10.396 (-0.51)	-9.738 (-0.61)	3.421 (0.36)	152.445 (3.50)	8.844 (0.40)
Panel B: tfidf Weights							
Fraud	0.001 (1.56)	0.003 (2.85)	0.006 (1.69)	0.012 (2.43)	0.005 (3.34)	0.057 (1.11)	0.010 (0.39)
Material weakness	0.004 (4.45)	0.004 (5.10)	0.012 (3.94)	0.014 (2.97)	0.006 (3.56)	0.153 (3.63)	0.041 (1.65)

observations. For the fraud results in Panel A, using simple proportional measures, none of the seven word lists have a significant coefficient after controlling

Table VII: fraud and material weakness logit regressions

Standardized Unexpected Earnings Regressions

The dependent variable is the earnings surprise, based on analysts' estimates (standardized by price and expressed as a percentage), for the quarterly number reported within 3 months of the 10-K file date. The coefficients are based on 60 quarterly Fama-MacBeth (1973) regressions with Newey-West (1987) standard errors using one lag. Each coefficient reported is from a regression also containing Fama-French (1997) industry dummies (based on 48 industries) and a constant. In addition, this regression contains Analyst dispersion, which is the standard deviation of analysts' forecasts in the most recent period prior to the announcement scaled by stock price in the month prior to the announcement. Analyst revisions are calculated using the mean forecast from the 4 months prior to the earnings announcement to compute the forecast revision. See the Appendix for the other variable definitions. The estimates are based on a sample of 28,679 10-Ks over 1994 to 2008. The coefficients for the word lists in columns (3) and (4) are multiplied by 100.

	Proportional Weights		tfidf Weights	
	(1)	(2)	(3)	(4)
<i>Word Lists</i>				
H4N-Inf	1.937 (2.58)		0.035 (4.03)	
Fin-Neg		2.683 (2.41)		0.030 (2.87)
<i>Control Variables</i>				
Log(size)	0.012 (3.22)	0.011 (2.93)	0.011 (2.94)	0.011 (2.90)
Log(book-to-market)	0.021 (2.23)	0.021 (2.20)	0.022 (2.27)	0.022 (2.27)
Log(share turnover)	0.001 (0.18)	0.000 (-0.02)	-0.002 (-0.23)	-0.002 (-0.21)
Pre.FFAlpha	16.913 (2.65)	16.646 (2.61)	16.520 (2.59)	16.630 (2.60)
Institutional ownership	-0.035 (-1.65)	-0.031 (-1.48)	-0.028 (-1.33)	-0.028 (-1.36)
NASDAQ dummy	-0.005 (-0.35)	-0.006 (-0.43)	-0.004 (-0.34)	-0.003 (-0.27)
Analyst dispersion	-45.218 (-5.93)	-45.515 (-5.99)	-45.647 (-6.01)	-45.643 (-6.02)
Analyst revisions	11.403 (4.45)	11.460 (4.47)	11.434 (4.48)	11.447 (4.49)
Average R^2	18.38%	18.54%	18.56%	18.53%

Table VIII: standardized unexpected earnings regressions

6 Short summary

- Generic sentiment dictionaries misclassify common financial words.
- Almost three-fourths of Harvard negative word counts in 10-Ks are not typically negative in financial contexts.
- The authors build finance-specific negative, positive, uncertainty, litigious, strong modal, and weak modal dictionaries.
- The Fin-Neg list is more closely related to filing-period returns than the Harvard negative list when proportional counts are used.
- TF-IDF weighting mitigates the Harvard list's misclassification problem by downweighting common words.
- The MD&A section does not produce a clearly stronger tone signal than the full 10-K.
- Uncertainty and litigious word lists are strongly related to trading volume and return volatility.
- Textual measures are also linked to fraud, internal-control weaknesses, and standardized unexpected earnings.

- The central lesson is that financial textual analysis requires finance-specific dictionaries and careful weighting.

7 Conclusion

7.1 Core conclusion

Loughran and McDonald show that word lists developed outside finance can be misleading when applied to 10-Ks. The Harvard negative list classifies many ordinary financial terms as negative, adding noise and sometimes bias to tone measures.

7.2 Economic intuition

Financial reporting has its own vocabulary. Words such as *liability*, *tax*, *cost*, or *capital* can be neutral accounting terms rather than negative sentiment. A finance-specific dictionary better separates true bad-news language from ordinary disclosure language.

7.3 Takeaway

The paper is a foundational methodological contribution to financial textual analysis. Its main message is simple: dictionaries are not portable across domains without validation. For 10-Ks and other financial texts, finance-specific word lists and term weighting are essential for credible measurement.